

Detecting Oral Cancer Using Images

ABSTRACT

Oral cancer remains a significant global health challenge, with early detection critical for improving patient outcomes. This study investigates the application of deep learning models to enhance the accuracy and efficiency of oral cancer detection using medical images. A dataset of 950 images, comprising 500 oral cancer and 450 non-cancer images, was preprocessed through normalization and augmented with techniques such as rotation, flipping, and scaling to improve model generalization. Four models were evaluated: a baseline Convolutional Neural Network (CNN), a CNN with data augmentation, ResNet50, and MobileNetV2.

The models were assessed using metrics such as accuracy, loss curves, confusion matrices, and classification reports. The results indicate that MobileNetV2 outperformed other models, achieving a test accuracy of 92.86%, with minimal false positives and false negatives. While the baseline CNN showed high training accuracy, it suffered from overfitting. ResNet50 struggled due to its complexity relative to the dataset size, highlighting the need for larger datasets or transfer learning for deeper architectures. Data augmentation proved crucial in improving the robustness of CNN models.

This study underscores the potential of lightweight architectures like MobileNetV2 in medical imaging applications, offering a promising approach for early and non-invasive oral cancer detection. Future work will focus on incorporating advanced regularization techniques, expanding the dataset, and leveraging transfer learning to further enhance model performance.

INTRODUCTION

Oral cancer is among the most common malignancies, with significant morbidity and mortality rates globally. Early detection is crucial, as late-stage diagnoses often lead to poor prognoses and limited treatment options. Current diagnostic methods primarily rely on clinical examinations and biopsies, which can be subjective, invasive, and resource-intensive. The rapid advancements in artificial intelligence (AI) and machine learning (ML) present a unique opportunity to develop efficient, non-invasive diagnostic tools to assist clinicians in early detection.

Deep learning, a subset of ML, has shown remarkable success in image-based tasks, including medical imaging. By leveraging convolutional neural networks (CNNs) and advanced architectures like ResNet and MobileNet, researchers have demonstrated improved performance in tasks ranging from image classification to object detection. However, in the domain of oral cancer detection, the potential of these techniques remains underexplored. This study aims to bridge this gap by evaluating the performance of multiple deep-learning models in classifying oral and non-oral cancer images..

PROBLEM STATEMENT

The manual diagnosis of oral cancer using traditional methods poses several challenges, including variability in expertise among clinicians, limited access to specialized

equipment in resource-constrained settings, and delayed detection, which significantly impacts patient outcomes.

While existing machine learning models have shown promise in cancer diagnosis, there remains a lack of consensus on the optimal deep learning architecture for oral cancer classification. Factors such as the choice of model, preprocessing techniques, and data augmentation strategies critically influence the reliability and robustness of the system.

This research addresses the need for a comprehensive evaluation of deep learning models for oral cancer detection. By systematically comparing multiple architectures and preprocessing strategies, we aim to identify an effective and efficient solution that can augment clinical decision-making, reduce diagnostic delays, and improve patient care outcomes.

SOLUTION

To address the challenges of early and accurate oral cancer detection, this study proposes a systematic evaluation of multiple deep learning models, leveraging modern convolutional neural network (CNN) architectures and preprocessing techniques. The proposed solution involves the following steps:

A. Dataset Preparation:

A curated dataset comprising images of oral cancer and non-oral cancer is utilized. The dataset is preprocessed to enhance image quality, normalize dimensions, and address class imbalances using data augmentation techniques such as rotation, flipping, and scaling.

B. Model Development:

Four models are implemented and evaluated:

- A baseline CNN without data augmentation, serving as a control model.
- A CNN with data augmentation to analyze its impact on performance.
- ResNet50, a deep residual network known for its superior feature extraction capabilities.
- MobileNetV2, a lightweight yet powerful model optimized for mobile and edge devices.

The models are implemented using a Python-based deep learning framework (TensorFlow) with GPU acceleration to handle computational demands efficiently.

C. Evaluation Metrics:

1)Model Performance:

Training Accuracy –The model's performance on the dataset used to train it.

Validation Accuracy – The model's performance on unseen data during training for tuning.

Test Accuracy – The model's performance on entirely new data for final evaluation.

2)Model Accuracy and Model Loss:

Model accuracy and loss curves indicate the model's learning progress, overfitting risk, and generalization performance across training, validation, and testing.

3)Confusion Matrix:

True Negatives: Correctly predicted non-cancer cases

False Positives: Non-cancer cases incorrectly predicted as cancer

False Negatives: Cancer cases incorrectly predicted as non-cancer

True Positives: Correctly predicted cancer cases

2)Classification Report:

The classification report evaluates the following :

- Precision -Indicates the proportion of true positive predictions out of all positive predictions.
- Recall (Sensitivity) - Measures the ability of the model to detect all actual positives.
- F1-Score - The harmonic mean of precision and recall
- Support - Number of instances for each class in the test data

DATASET

The dataset used for this study is the *Oral Cancer Dataset 2.0*, which comprises a total of 950 images: 500 oral cancer images and 450 non-cancer oral images. This dataset is curated with the aim of facilitating research and development in the field of oral cancer detection using machine learning algorithms. It contains images of varying quality, which represent the diversity of oral cancer lesions, as well as healthy oral tissue, offering a balanced representation for training and evaluation purposes.

Each image in the dataset is meticulously labeled to ensure proper classification. The dataset has been preprocessed to improve image quality, ensuring consistency in the data fed into machine learning algorithms.

Preprocessing steps include the normalization of image dimensions, ensuring uniformity across the dataset for model training. Additionally, the dataset is subjected to various data augmentation techniques, such as rotation, flipping, and scaling, in order to simulate different real-world conditions and to address potential class imbalances. These augmentation strategies are particularly important, as they provide an expanded and diverse set of images, improving the generalization ability of the models.

EVALUATION

The models are implemented and evaluated are:

1) CNN without Augmentation:

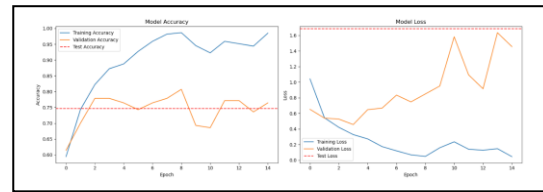
- Model Performance:

Training Accuracy – 98.48%

Validation Accuracy – 76.43 %

Test Accuracy – 74.65 %

- Model Accuracy and Model loss:



- Confusion Matrix :

True Negatives: 49 False Positives: 25

False Negatives: 11 True Positives: 57

- Classification Report :

	precision	recall	f1-score	support
Non-Cancer	0.82	0.66	0.73	74
Cancer	0.70	0.84	0.76	68
accuracy			0.75	142
macro avg	0.76	0.75	0.75	142
weighted avg	0.76	0.75	0.75	142

Warning : The model might be overfitting . Need to use regularization techniques .

2) CNN with Data Augmentation:

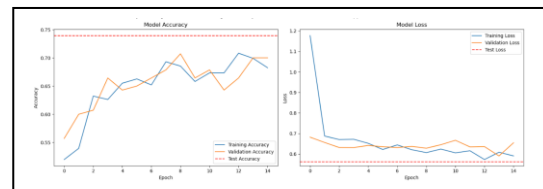
- Model Performance:

Training Accuracy – 68.24%

Validation Accuracy – 70.00 %

Test Accuracy – 73.94 %

- Model Accuracy and Model loss:



- Confusion Matrix :

True Negatives: 49 False Positives: 25

False Negatives: 12 True Positives: 56

- Classification Report :

	precision	recall	f1-score	support
Non-Cancer	0.80	0.66	0.73	74
Cancer	0.69	0.82	0.75	68
accuracy			0.74	142
macro avg	0.75	0.74	0.74	142
weighted avg	0.75	0.74	0.74	142

3) ResNet50:

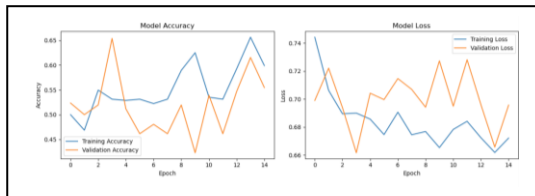
- Model Performance:

Training Accuracy – 65.20%

Validation Accuracy – 54.96 %

Test Accuracy – 57.14 %

- Model Accuracy and Model loss:

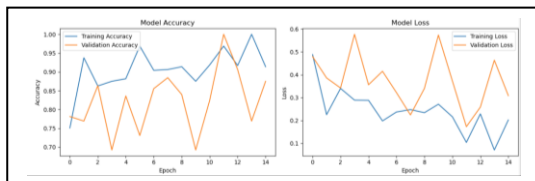


- Confusion Matrix :
True Negatives: 60 False Positives: 13
False Negatives: 47 True Positives: 20
- Classification Report :

	precision	recall	f1-score	support
Non-Cancer	0.56	0.82	0.67	73
Cancer	0.61	0.30	0.40	67
accuracy			0.57	140
macro avg	0.58	0.56	0.53	140
weighted avg	0.58	0.57	0.54	140

4) MobileNetV2:

- Model Performance:
Training Accuracy – 93.47 %
Validation Accuracy – 85.11 %
Test Accuracy – 92.86 %
- Model Accuracy and Model loss:



- Confusion Matrix :
True Negatives: 70 False Positives: 3
False Negatives: 7 True Positives: 60
- Classification Report :

	precision	recall	f1-score	support
Non-Cancer	0.91	0.96	0.93	73
Cancer	0.95	0.90	0.92	67
accuracy			0.93	140
macro avg	0.93	0.93	0.93	140
weighted avg	0.93	0.93	0.93	140

Warning : The model might be overfitting . Need to use regularization techniques .

CONCLUSION

The research focused on detecting oral cancer using a deep learning-based image classification approach, implementing and evaluating four models. The results provided significant insights into the performance and limitations of these models, as summarized below:

A) CNN without Augmentation

The model demonstrated excellent performance during training. However, its validation and test accuracy were significantly lower, indicating overfitting. The confusion matrix and classification report highlighted that while the model could detect oral cancer cases to a reasonable extent, it also struggled with false positives and false negatives. This suggests that the lack of data augmentation and regularization techniques adversely affected the generalization capability of this model.

B) CNN with Data Augmentation

Introducing data augmentation led to more balanced performance, with a reduction in overfitting. The training accuracy dropped, but the validation and test accuracies showed improved. The confusion matrix indicated slightly better results for detecting cancer cases compared to the non-augmented CNN. Although the model did not achieve the highest accuracy, data augmentation proved to be an essential strategy for improving generalization and robustness.

C) ResNet50

ResNet50, a deeper architecture, underperformed relative to expectations. The confusion matrix revealed that the model struggled significantly with false negatives, failing to detect a considerable number of oral cancer cases. This underperformance may be attributed to insufficient data for training such a deep model effectively or inadequate hyperparameter tuning. ResNet50's poor generalization suggests that deeper architectures may require either larger datasets or advanced techniques like transfer learning tailored for the specific problem.

D) MobileNetV2

MobileNetV2 emerged as the best-performing model. The confusion matrix demonstrated its superior ability to accurately detect both cancer and non-cancer cases, with minimal false positives and false negatives. MobileNetV2's lightweight architecture combined with efficient feature extraction likely contributed to its robustness and high accuracy, making it highly suitable for oral cancer detection.

Future Directions

- Regularization and Optimization: Incorporating techniques such as dropout, L2 regularization, and advanced optimization algorithms could further enhance performance and reduce overfitting.
- Transfer Learning: Applying pre-trained weights for deeper architectures like ResNet50 could address data limitations and improve model performance.
- Dataset Expansion: Expanding the dataset with more diverse and high-quality oral cancer images would benefit all models, especially deeper architectures.

In conclusion, MobileNetV2 proved to be the most reliable and efficient model for oral cancer detection in this study,

while data augmentation and lightweight architectures emerged as critical factors for success in medical image analysis.